

KOUVOLA ANJALA, Finland

WMO: 028300

Lat: 60.696N

Lon: 26.811E

Elev: 33

StdP: 100.93

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-19.8	-24.9	0.4	-22.6	-21.7	0.5	-19.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	27.9	19.5	26.0	18.2	24.0	17.2	20.9	25.5	19.6	24.1	18.4	22.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.0	23.2	17.9	12.9	21.9	16.7	12.0	20.2	60.4	25.5	56.2	24.3	52.2	22.5	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-24.0	29.5	4.8	2.5	-27.5	31.3	-30.3	32.8	-33.0	34.2	-36.6	36.0		
(5)			-24.2	21.7	4.8	1.6	-27.7	22.9	-30.5	23.9	-33.2	24.8	-36.7	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-6.9	-5.0	-1.7	4.4	11.0	14.8	17.9	16.1	11.5	5.1	1.6	-2.5
	DBStd	9.32	6.77	6.00	4.32	3.42	4.23	3.40	3.32	2.99	3.20	3.50	4.14	5.53
	HDD10.0	2338	522	419	363	173	38	4	0	1	22	154	253	388
	HDD18.3	4717	781	652	622	419	231	116	50	82	205	410	503	646
	CDD10.0	721	0	0	0	4	68	147	244	189	67	3	0	0
	CDD18.3	59	0	0	0	0	2	8	36	12	0	0	0	0
	CDH23.3	549	0	0	0	0	38	105	319	87	0	0	0	0
	CDH26.7	107	0	0	0	0	4	14	73	16	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	650	50	42	37	31	41	60	62	71	63	69	66	59
	PrecMax	908	91	115	71	73	104	117	123	141	147	150	143	166
	PrecMin	453	9	2	2	8	9	9	8	5	14	18	8	13
	PrecStd	102	22	25	18	16	25	26	29	33	35	34	28	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.5	9.7	20.2	27.1	28.3	30.8	28.6	21.4	14.0	10.4	8.2
		MCWB	3.5	2.9	5.1	11.2	15.9	18.1	20.9	20.0	16.6	11.4	9.2	7.7
	2%	DB	3.1	3.3	6.9	16.9	24.0	26.2	28.8	25.7	19.4	12.4	9.0	5.9
		MCWB	2.7	2.1	3.4	9.3	14.8	17.3	20.5	18.6	15.4	10.8	8.5	5.3
	5%	DB	2.2	2.3	4.8	13.5	21.9	24.2	26.9	23.5	17.9	11.5	7.6	4.6
		MCWB	1.9	1.7	2.4	7.6	13.7	16.7	19.4	17.6	14.4	10.2	7.1	4.1
	10%	DB	1.1	1.5	3.3	10.9	19.5	22.1	24.9	21.9	16.6	10.4	6.7	3.3
		MCWB	0.8	1.1	1.6	6.0	12.3	15.3	18.3	16.5	14.1	9.4	6.3	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.2	5.5	11.5	17.9	20.3	22.5	22.0	17.5	12.6	9.5	7.9
		MCDB	3.9	4.4	8.4	17.9	24.0	26.0	27.7	26.9	19.9	13.4	10.3	8.3
	2%	WB	2.8	2.5	3.9	10.0	16.2	18.7	21.5	19.9	16.4	11.6	8.3	5.4
		MCDB	3.0	3.2	5.9	15.4	21.4	23.7	26.8	23.7	18.2	12.1	8.8	5.8
	5%	WB	1.9	1.8	2.9	8.2	14.7	17.2	20.4	18.6	15.5	10.6	7.2	4.1
		MCDB	2.2	2.2	4.3	12.3	20.0	22.4	25.1	22.0	17.1	11.2	7.5	4.5
	10%	WB	0.9	1.1	2.0	6.8	13.2	16.0	19.3	17.5	14.5	9.5	6.3	2.9
		MCDB	1.0	1.5	3.1	10.0	18.4	20.8	23.7	20.5	16.1	10.2	6.6	3.3
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.4	7.3	9.4	11.2	10.0	9.8	9.2	7.6	5.9	3.8	4.1
		MCDBR	3.5	4.4	8.6	14.6	14.7	13.7	12.6	11.9	10.0	7.1	4.5	4.4
		MCWBR	3.3	3.8	5.7	8.1	7.0	6.3	5.4	5.9	6.2	5.4	4.3	4.4
	5% WB	MCDBR	3.7	3.9	6.9	12.6	12.9	11.8	11.1	9.8	8.1	6.1	4.3	4.3
		MCWBR	3.6	3.5	4.8	7.6	6.6	6.1	5.2	5.3	5.4	5.2	4.2	4.4
Clear-Sky Solar Irradiance	taub	0.271	0.308	0.331	0.369	0.367	0.363	0.385	0.375	0.349	0.330	0.296	0.376	
	taud	2.120	2.169	2.230	2.311	2.405	2.451	2.409	2.431	2.488	2.393	2.197	2.592	
	Ebn at Noon	460	663	784	824	858	867	835	813	773	649	426	316	
	Edh at Noon	41	72	94	105	104	101	103	93	75	59	39	27	
All-Sky Solar Radiation	RadAvg	0.25	0.86	2.35	3.84	5.23	5.46	5.45	4.16	2.45	1.02	0.27	0.13	
	RadStd	0.05	0.18	0.29	0.41	0.57	0.35	0.50	0.44	0.29	0.15	0.05	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	-160	-172	N/A	N/A	

Nomenclature: See separate page